Point-of-Interest Recommender Systems

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Chapter 1

Introduction

In this chapter, we give a brief introduction of the research in this thesis, including the background, definitions, and problem characterization.

1.1 Trajectory Data Mining

With the rapid development of technology and portable devices in our daily life, the velocity and volume of data have been affected. Moreover, the growth and progress in location-based and location acquisition technologies have caused a plenty of spatio-temporal trajectories demonstrating the mobility of different moving objects, such as people, vehicles, and animals [1]. A trajectory is a footprint generated by moving objects which determines the location and time in geographical and temporal spaces of that, generally shown as a series of chronologically ordered points, e.g. $P_1 \rightarrow P_2 \rightarrow P_3 \rightarrow ... \rightarrow P_n$, which each point comprises of a geospatial coordinate set and a timestamp such as $p(x, y, t)$ [1, 2, 3]. Such trajectories bring new opportunities and challenges, offer us unexampled information to mining, and understand moving objects. Spatio-temporal data mine is a key technique in research and improves practical applications such as urban planning, relieving traffic congestion, effective location recommendation systems, and intelligent transportation systems [1, 2]. The methods and algorithms in classical data mining can be used in trajectory data mining or spatio-temporal data mining but such information (i.e. spatial and temporal) also makes it possible to think and examine new and novel approaches for analyzing data in the various fields of spatio-temporal data mining [4].

1.2 Location-Based Social Networks

Recent years, social networks like Facebook[1], Twitter[2], WhatsApp[3], and Instagram[4] with millions or even billions of users have become some of the most used and popular internet-based applications. A social network is an abstract structure and a digital representation of our real world, as we have different relationships in our personal life, so it contains different relations between the individuals, e.g. friendships, common interest, and shared knowledge [5]. In addition, by sharing ideas, activities, events, news, and interests through an online platform the user social connections grow [5, 6]. The addition of other aspects in social networks strengthens the connection between the social networking services and the real-world social networks. One of the most important aspects is location or geotag, which has led to the formation of Location-Based Social Networks (LBSNs) [5]. LBSNs or Geosocial Networks not only adds the location as a new aspect in social networks, but also adds a new structure and formally defines as a type of social network which contains dependency between locations, mediums, and users in the real world [6]. In LBSNs such as Foursquare[5], GeoLife, Gowalla[6], and Whrrl[7], users can easily share their experiment in geo-spatial locations and location-related contents in the physical world via such online platforms. For example, when a user goes to a restaurant he or she can easily share comments, images, and the other media with his or her friends [5]. An overview of the different types of networks contained in a typical LBSN shown in Fig. 1.1,

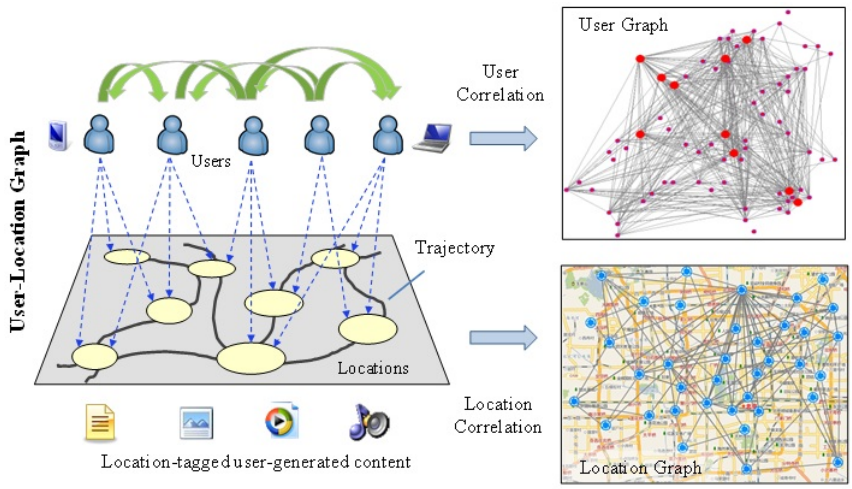

Figure 1.1: Concept of location-based social networks [5]

[1]https://www.facebook.com/
[2]https://twitter.com
[3]https://www.whatsapp.com/
[4]https://www.instagram.com/
[5]https://www.foursquare.com/
[6]https://en.wikipedia.org/wiki/Gowalla
[7]https://en.wikipedia.org/wiki/Whrrl

1.3 Recommender Systems

Nowadays, with the growth of data and rapid development of online services such as Facebook, Twitter, YouTube[1], Spotify[2], and Netflix[3] the main challenge for people is overwhelming choices over the internet and selection of content in the online applications.

This problem is known as the information overload problem and it becomes more and more serious [7]. Therefore, the effective selection of items that fit user's preferences and interests becomes more and more challenging for their. For example, as there are hundreds of thousands of movies in the Netflix, pieces of music in the Spotify, videos in the YouTube, and the other media in Twitter and Facebook, it will be very difficult for a specific user to search and find interesting contents. To address and overcome this problem, recommender systems are an effective and promising way [6]. Recommender systems as an important and essential approach for internet and online application have a great impact on user or customer satisfaction. Such satisfaction can happen by automatically suggesting items of interest based on the user's past behaviors and interests as well as feedbacks (e.g. like or dislike). Fig. 1.2 shows an overview of how a recommender system works.

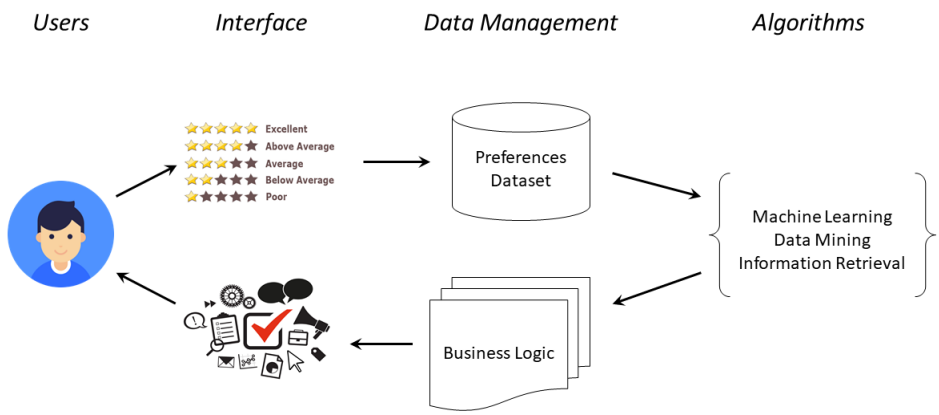

Figure 1.2: An overview of how to a recommender systems

1.4 Recommendation Tasks

In the recommendation, there are two basic and important tasks that how to recommend an item to a user [7]:

- **Rating prediction**: This task refers to when the system generates a prediction for an active user and a given item, for example, to predict how many stars a user may give to a

[1]https://www.youtube.com/
[2]https://www.spotify.com/
[3]https://www.netflix.com/

movie or music.

- **Top-N item recommendation**: Top-N task try to return a list of interesting items. Actually, it is to select and recommend the top-N items upon which the user is likely to act (e.g., watch, purchase, read). For instance, recommend a user the top-10 most interesting items along with their ranking order in the recommendation list.

1.5 Recommendation Methods

In general, a recommendation problem can be formulated as follow. There are two important sets, U as a set of all users and I is a set for all candidate items, e.g. movies, music, books, jobs etc. To calculate the effect of recommending item i to user u a utility function f is used (i.e. $f : U \times I \rightarrow R$). For each user $u \in U$, item $i^{'} \in I$ is chosen to maximize the user's utility. Formally, we have:

$$\forall u \in U, i_u^{'} = arg\ max_{i \in I}\ f(u,i) \tag{1.1}$$

To this purpose, the techniques of recommender systems classified into three categories, *Collaborative Filtering*, *Content-Based Method*, and *Hybrid Models* [5, 6], where each method is described below:

1.5.1 Collaborative Filtering Recommendation

The basic idea behind the collaborative filtering (CF) method, which is known as one of the most effective methods to predict user's interest, is to aggregate preferences of like-minded user, similarity of like-rated items, or to learn a preference pattern from historical rating and interesting items. Therefore, the main input in CF approaches is the User-Item Rating Matrix. For example, Fig. 1.3 shows the user-item rating matrix of users for movies. Based on this idea, the CF approaches can be subdivided into *Memory-Based* and *Model-Based* approaches. The previous works show the model-based approaches outperform the memory-based approaches in terms of predictive accuracy and scalability [6, 7, 8].

The memory-based approaches, which has been considered in many traditional and commercial recommender systems, use the entire of user-item rating matrix for the recommendation. The memory-based approaches are divided into *User-based* and *Item-based* methods according to which similarity (users or item) has been made to the recommendation:

- **User-based**: The user-based method finds similar users based on their ratings on items and according to this similarity recommends items to users [6]. For example, in a LBSN such as Foursquare, considers the most similar users to a target user based on the previously visited locations and recommends new locations to the target user.

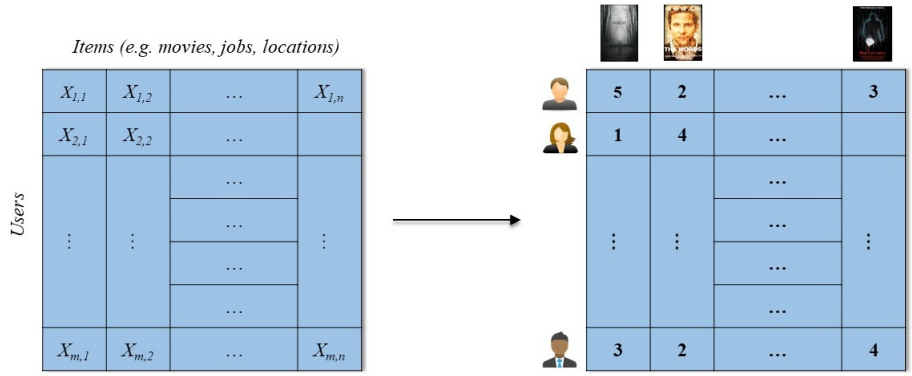

Figure 1.3: User-Item rating matrix

- **Item-based**: This method finds the similarity between items according to users rating on different items and recommends items based on such similarity [6]. For instance, if users often rate a drama movie, such movies are recommended to them.

Three distance similarity metrics are usually use in CF, Jaccard similarity, Cosine similarity, and Pearson similarity.

Memory-Based CF approaches are very simple and easy to adopt because they are well-understood. However, there are some problems and shortfalls in the memory-based CF approaches [6, 8]:

- **Cold Start**: The cold-start problem appears when a new user or item added to the system because they have rarely information in user-item rating matrix. Therefore, it is difficult for them to be recommended.

- **Sparsity**: As there are a lot of users and items, the user-item rating matrix usually is sparse. According to [8] in many real world applications, the sparsity is the density of 10^{-4} to 10^{-5}. When the data is sparse, the similarity measure may not be reliable.

- **Scalability**: Memory-based approaches consider the whole user-item matrix to calculate the similarity and perform recommendation. This process needs a large storage space. In addition, to computing the similarity between users and items has high computational costs.

To address the above issues, model-based CF is proposed, which are based on a learning model. Hence, they use data mining and machine learning algorithms and technologies to make a model and train it according to the training data. Then, this model can be used to predict ratings and recommends to the users. Various models investigated in this category and applied to the different recommender systems. For example, latent factor models, classification/regression

models etc. Among them, latent factor models such as the Matrix Factorization (MF) model have been widely used and it is more effective in the accuracy and scalability [8].

MF, as the Netflix Prize competition has demonstrated, is one of the efficient and the most popular method of model-based CF. MF models outperform the classic memory-based techniques for generating rating and recommendation [7, 9, 10, 11]. This method has been proved the most accurate method to alleviate the problems of memory-based methods [11]. In addition, they are able to handle large-scale datasets [7]. The basic idea behind MF is to take into account latent factors to both of item's properties and user's interests [7, 8]. For example, in a movie recommendation, genre, director, the number of actors, music etc., are the latent factors. As shown in Fig. 1.4, MF decomposes the user-item rating matrix $R_{m\times n}$ into two low-rank user-factors matrix $P \in \mathbb{R}^{m\times k}$ and item-factors matrix $Q \in \mathbb{R}^{n\times k}$, where k is the number of latent features. Then the rating prediction of a user u on an item i is calculated by the inner product between the user's vector and the item's vector as below [7, 8]:

$$\acute{r}_{u,i} = P_u^T Q_i \tag{1.2}$$

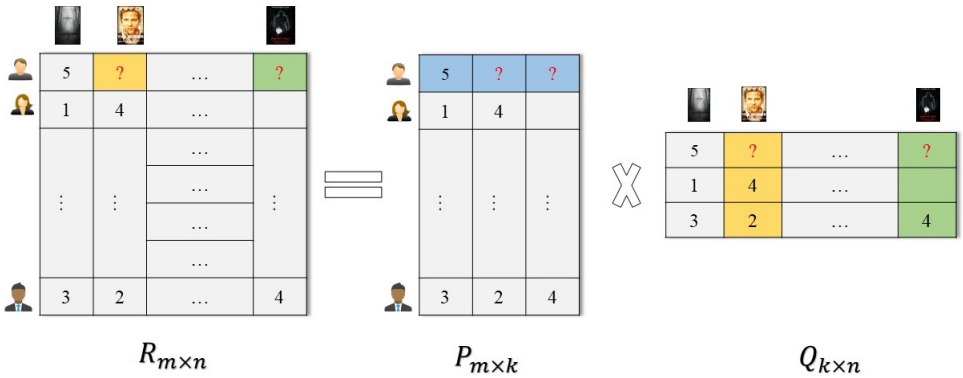

Figure 1.4: Matrix Factorization model

1.5.2 Content-Based Recommendation

Content-based (CB) approaches consider the user information to make a recommendation. As shown in Fig. 1.5, it uses user's historical data and actions in the system as well as user's profile when he or she completes them in the registration process (e.g., age, gender, and preferred cuisines). One of the main problem in CB recommendation is the lack of data diversity, as it recommends the items that are similar to the user's interest. Moreover, if a user does not complete his or her information in the registration process it will suffer from the cold-start problem as well[5, 6, 8].

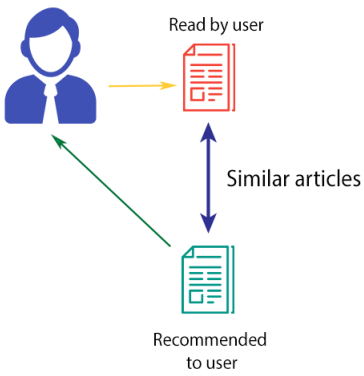

Figure 1.5: An illustrate of content-based approach [12]

1.5.3 Hybrid Model

According to the advantages and disadvantages of CF and CB approaches, hybrid approaches make use of the advantages and combine CF and CB recommender systems to recommend items. The fundamental assumption avoids the disadvantages of CB and CF, for instance, use CB when a cold-start problem may happen [6, 8]. Fig. 1.6 shows an illustration of hybrid models. Also, Fig. 1.7 demonastrates a summary of recommendation approaches.

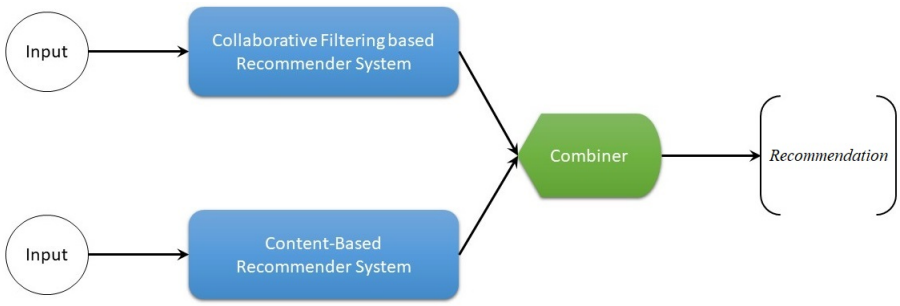

Figure 1.6: Hybrid model

1.6 Point-of-Interest Recommendation

With the rapid growth of Web 2.0, wireless communication technologies, and global position system (GPS) and also with the increase of smart phones and portable devices, LBSNs have become very popular and attract lots of attention from people as well as industry and academia [13, 14, 15]. According to [13], many people (i.e. about 52%) in the US use LBSNs. Although LBSNs provide social interactions for users, the more interested in LBSNs is to make use of the rich information to mine user's preferences on locations and recommend new places where users may be interested in. In LBSNs, users can connect with their friends, share comments,

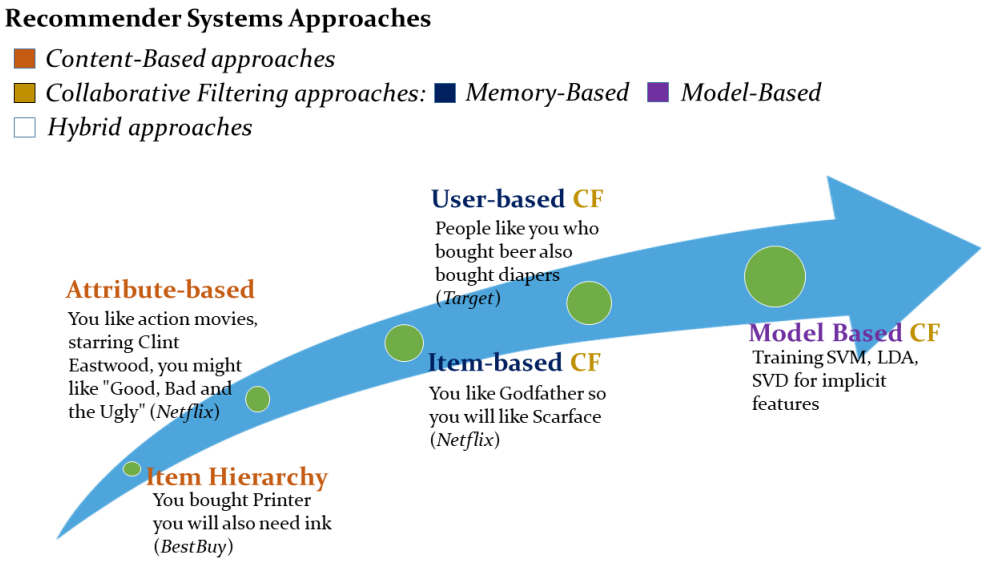

Figure 1.7: A summary of recommendation approaches

photos, and their location via check-in in a Point-of-Interest (POI) location such as restaurants, cafes, stores etc., to share life experience in the physical world. Based on [15], Foursquare had more than 8 billion check-ins, 65 million venues, and over 55 million users in the world until June 2016. To improve the user experience, the task of POI recommendation as a location-based recommender system recommends new interesting locations to users. POI recommender systems are useful and can be used for both users and the location provider. Besides recommending a new location to users, they also help to advertise agencies in advertising based on customers, such as a location-aware advertisement [14]. Therefore, users can easily search nearby restaurants or downtown shopping and other POIs are able to aware users to easily find them. In POI recommendation, the first valuable information is the user's check-in records that indicate the user's presence in a location (POI). In addition, there is other important content information in LBSNs. This information answers some basic questions in LBSNs (i.e. Who, When, Where, What) [8, 13]. As shown in Fig. 1.8, when a user check-in in a special location, this check-in not only shows user's interest on a location but also consists of the contextual information such as location information (i.e. location category, latitude and longitude), user's social ties, check-in time stamps, and check-in related content (e.g. tips, comments, POI descriptions, etc.).

The aforementioned information that is based on check-in data, answers the basic questions in LBSNs. For example, the check-in time stamp indicates when a user checked-in a location. This information is also called as *LBSN Context* [8, 13, 14, 15].

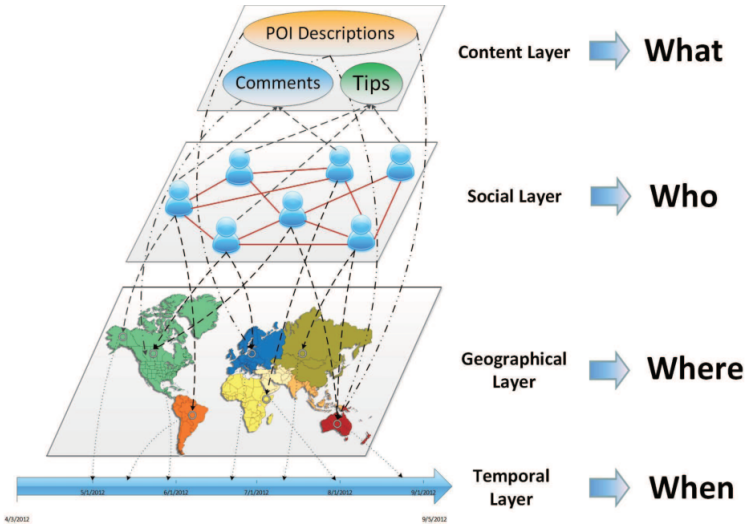

Figure 1.8: The Information Layout of LBSNs [8]

1.7 Context-Aware Recommendation

In recent years, recommender systems have been considered for different recommendations in various domains such as books, movies, dating, and items. Though traditional techniques are effective but in personalized POI recommendation, the heterogeneous information layout and the specificity of human mobility have a significant impact on the result, since they bridge the gap between social and physical world. Therefore, designing an efficient POI recommendation approache on LBSNs, necessarily needs consider the following properties based on different layer information:

1.7.1 Geographical Information

Each location or POI has two important attributes latitude and longitude that show the location of the POI. The physical location of users has a significant influence on the locations that users will visit [16]. Based on Tobler's First Law of Geography, *Everything is related to everything else, but near things are more related than distant things (Tobler 1970)*. In LBSNs users prefer to visit nearby locations rather than distant ones and users may be interested in POIs around a POI that users have checked-in [14]. Therefore, geographical distance is one the most significant context information in POI recommendation to show user's check-in behavior, and it is the most factor to differentiate between POI recommendation and traditional recommender systems. According to the physical location of users and the POI locations, the user's check-ins distribution can be considered.

In particular, two representative models, i.e., Power-law distribution model and Gaussian distribution model, are the best proposed model to capture the geographical influence in POI

recommendation.

1.7.2 Temporal Information

Time is another important attribute in the check-ins and the user's behavior. The temporal information shows the user's habit or user's interest drift. The geographical movement of users is strongly relevant to the location properties and two user's behavior can be different over time [8, 16, 17]. Additionally, the user's check-in frequency in a POI over time shows the user's interest for that location.

For example, on weekdays people visit places around their home and office, which is really different from the weekends that users spend time in shopping malls and restaurants. Therefore, user's check-ins strongly diverse based on years, months, weekdays, and time in a day (i.e. lunchtime, midnight etc.) [15]. Such temporal information follows a periodic pattern. On the other hand, temporal consecutive pattern happens when a user's check-ins are ordered. For instance, a user may want to drink a cup of coffee at a coffee before he goes to the office. In addition, each POI has different opening hours and peak time (e.g. restaurants and bars), thus their check-ins are different over time [17].

Therefore, the temporal patterns help us to better understand the user's check-ins behavior and to design more advanced and accurate POI recommender systems on LBSNs.

1.7.3 Social Information

One the most common information in LBSNs is social information. Traditional recommender system methods use this information to make the recommendation as user-based CF approach. ÿåhe idea is that, the users share the common interests in LBSNs and suggest to their friends the locations, the social influence considers enhancing POI recommender methods [15, 16]. Previous studies have shown that the impact of social influence in POI recommendation is limited because most of the users share a few common visited interests [14, 15].

1.7.4 Categorical Information

Each POI is a special location that gives users a certain service. Therefore, each POI is indicated by a few categories. The category of a POI shows the activities happen in the POI and the products or services are provided by the POI [18]. For instance, a person visiting a restaurant means that she or he may have a meal there and a Chinese restaurant indicates that Chinese food will be provided for customers. In addition, the popularity of a POI reflects the quality of products or services offered by the POI, e.g., a popular shopping usually indicates that it supplies high quality products. Therefore, it is helpful to utilize the category information for POI recommendations.

1.7.5 Content Information

Recent studies show researchers incorporate the content information in LBSNs such as tips, ratings for POIs, photos, media, and comments to improve the accuracy of recommendation as well as to address the data sparsity [13, 15]. For instance, user comments enable us to explore the user interest on the location as the check-in does not indicates the user likes the POI exactly.

1.8 Contextual Information Summary

Fig. 1.9 shows a summary of contextual information in POI recommendation and the most used and effective models for them.

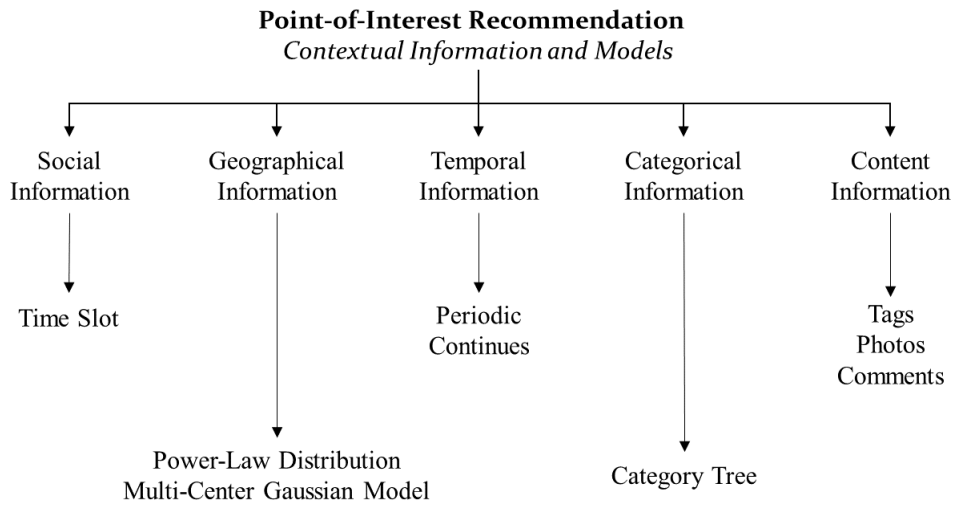

Figure 1.9: Contextual information summary

1.9 Problem Statement

In this review, we focus on the problem of POI recommendation by mining the user's historical behavior data in LBSNs. For ease of presentation, the key data structures and notations used in this research are summarized in Table 1.1.

Definition 1 (POI) *A POI is a geographical point with specific functions (e.g., hotel, restaurant, museum, store) that user may find useful or interesting.*

Here, a POI has two attributes: *identifier*, *location*. We use p to represent a POI identifier, l_p to denote its corresponding location identifier. The location information available for each POI p

Table 1.1: Terms used in the POI recommendation

Terms	Definition
U, L, P	the set of users, locations and POIs
D_u	the profile of user u
$p_{u,i}$	the POI of ith record in D_u
$l_{u,i}$	the location of POI $l_{u,i}$
$t_{u,i}$	the time of ith check-in record in D_u

in the collected raw datasets is in the form of the (latitude, longitude) pair.

Definition 2 (User Activity) *A user activity is made of a four tuple* (u, p, l_p, t) *which indicates that the user* u *visits the POI* p*, located at* l_p *at the time* t*.*

Definition 3 (User Profile) For each user u, we create a user profile D_u, which is a set of user activities associated with u. The dataset D used in our model consists of user profiles, that is, $D = Du : u \in U$.

Then, given a dataset D as the union of a collection of user profiles, we aim to provide POI recommendation for users. We formulate our problem as follows:

Problem 1 (POI Recommendation) *Given a user activity dataset D, a target user u with his/her current location l and current time t (that is, the query is q = (u, l, t)), our goal is to recommend a list of POIs that u would be interested in and would like to visit in the future based on location-based social network context (e.g. geographical location of POIs, social connections, temporal information (i.e. timestamps of check-ins), content information of check-ins) related to him as shown in the Fig. 1.10:*

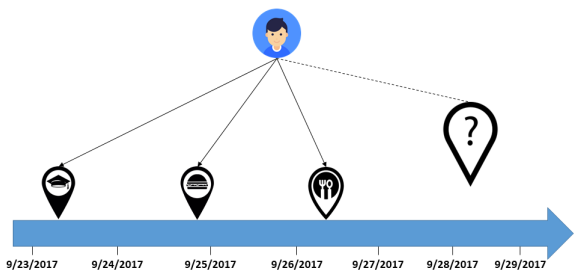

Figure 1.10: The Point-of-Interest recommendation

CHAPTER 2

LITERATURE REVIEW

In the last decade, recommender systems have been widely studied among various categories, e.g., movie recommendation on Netflix, job recommendation on LinkedIn, item recommendation on Amazon, news recommendation on Yahoo, friend recommendation on Facebook. POI recommendation also referred to as location recommendation, has been recognized as an essential task on recommender systems for enriching human life experience,targeted advertising, and facilitating decision making, which belongs to a sub-category of recommender systems. We first reviewed location-based social networks in Section 1.2, discussed recommender systems in Section 1.3. In this chapter, we review studies and techniques related to the location-based recommender systems for personalized POI recommendation. The related works categorized based on the contextual inforamtion.

2.1 Geographical Influence

In [19], a Ranking-Based Geographical Factorization Method (Rank-GeoFM) has been presented. To create a ranking model the authors used of check-in data frequency, in which higher frequency means the higher rank based on Bayesian Personalized Ranking (BPR). Based on the User's preference ranking from the check-in data and the recommendation score (calculated by Geographical Factorization model) they compute ranking incompatibility. This measure shows the number of POIs that are incorrectly ranked. Then, a fast learning schema considered to learn and compute the learning parameters. After that, they used temporal information as another context information to incorporate in the Rangk-GeoFM.

As the new model for geographical information, the [10] according to their studies on datasets, they find that user's check-ins are around several centers and each user has different numbers of centers. Therefore, they propose a Multi-Center Gaussian Model (MGM) to calculate geographical

influence. Then, in [20] Bayesian Personalized Ranking have been used to recommend a ranking list of POI because they considered top-k POI recommendation as User's preference.

2.2 Temporal Influence

As new intuition on user's check-in the paper [21] considered the check-in dataset in LBSNs indicate the confidence of users but no preference. Therefore, this property shows the implicit feedback dataset in LBSNs. Based on this intuition, they proposed a Weighted Matrix Factorization (WMF) method. Then, in [22] GeoMF that is a geographical model has been considered into the factorization model of WMF. The geographical idea in [22] is differentiate between unvisited but interesting POI among the negative ones for each user by spatial clustering. In [23] Temporal influence considered as a parameter to improve the negative modeling of POIs. Actually, they change the value of influence area by the time spent by a user to go from a POI i to the other POIs in grid l.

In [24], the authors proposed an Aggregated Temporal Tensor Factorization (ATTF) model that use three temporal features together, Periodicity, Consecutiveness, and non-uniformness. The authors of the previous papers proposed a temporal approach that they considered just periodic or continuous temporal features. They make use of Matrix Factorization but in this paper they use tensor to model the three-dimensional of temporal features. The non-uniformness means user's check-in preference changes at different hour in a day. They model user, time, and POI as a user-time-POI tensor then they used Pairwise Interaction Tensor Factorization (PITF) to decompose the user-time-POI tensor into the user-time, user-POI, and time-POI latent matrix. To inference the model they considered Bayesian Personalized Ranking (BPR).

In [25], the authors present a model based on user's behavior and similar behaviors among users. First of all, they propose a new data structure called behavior. It is a tuple of location category and time interval. Based on this data structure they generate a user-behavior bipartite graph that shows the relationships between users and behaviors. After that, they consider a Matrix Factorization model, in which the generated graph is the input of MF model. Moreover, they utilize the reachability of locations by different users as the spatial model. Finally, they compute the probability and recommend the location to users based on the two aforementioned model.

In [26], the authors try to consider a new model of temporal influence. Many previous works tried to model the correlation of temporal influence between users and locations. This paper model the user-specific and POI-specific temporal properties to show the temporal orientation of users and locations. They just consider periodic temporal information (i.e. weekday and weekend). The intuition can be when a user has more visit in weekend, so we should suggest the POIs that have more attract on weekends. By consideration of user's frequency in weekday or weekend and also POI's visit in weekday or weekend they make a naive model to improve the POI recommendation.

2.3 Social Influence

In [27], the authors present a hybrid model to location suggestion called SocialMix. SocialMix considers user's familiarity and preference similarity as a hybrid location suggestion model. As shown in Fig 2.1 for the first part, user's familiarity, they use three features to calculate the familiarity of users. They consider mutual friends, Cosine similarity, and Jaccard similarity as three feature variables to model user's familiarity then based on maximum likelihood estimation estimate the weight of each feature. For the second part, they calculate preference similarity by cosine similarity according to the user-location check-in matrix. Then, they combine user's familiarity and preference similarity to obtain the similarity between individuals. The model suggests location based on the similarity between users that calculated by the hybrid model.

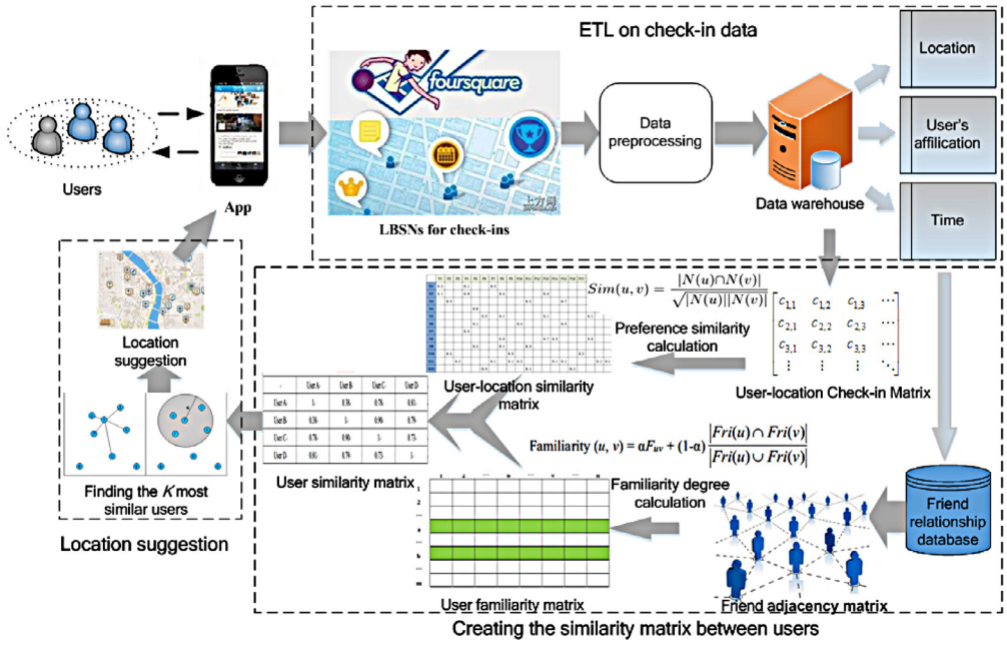

Figure 2.1: SocialMix Model [27]

One of the main challenges in POI recommendation is data sparsity. In [28], the authors try to address this problem by user's similarity. They consider 24-time slots then based on each user check-in in each time slot make a vector. So, they calculate the user similarity by cosine similarity measure. Their main idea is to divide a day into 24 time slots.

2.4 Category Influence

As each POI is a special location they are related to a category and in LBSNs typically organized by a hierarchical category tree. For instance, Foursquare offers a 3-level category hierarchy where top level consists of night life, food, etc., and lower level consists of bar, pub, Japanese food,

coffee, etc. Therefore, such category information indicates the semantic meaning of a location and reflects user preference on the corresponding category.

In [29], a category-aware POI recommendation has been considered. To this end, the authors considered a User's preference transition over the category of each location. First, they cluster the Users, to more accurate and address to sparsity, based on their Check-in behavior. To describe a user's check-in behavior, they considered Temporal Distribution (i.e. partition a day to *Working hours* and *Leisure hours* and a week to *weekdays* and *weekend*) of her/his check-in frequency in each category. Then, for each user cluster, they applied MF to predict the probability of User's Preference Transition based on the current POI category (choose $top-K1$ most likely categories). After that, category-aware POI recommendation perform to select $top-K2$ most likely POI in each category. Therefore, the result is a list of $K1 \times K2$ POIs for target user.

2.5 Content Influence

Content information such as comments, tag, decription etc on LBSNs can provide more information about a POI. For example, by observing a POI's description as "vegetarian restaurant", we infer that the restaurant serves "vegetarian food" and users who check-in at this POI might be interested in the vegetarian diet. This is an example of POI Properties. Moreover, the user's comments have benefit in sentiment analysis of user's check-in actions.

In [11], the authors present a model for POI recommendation incorporate three different influences. They use Multi-Tag, Social, and Geographical influence with Matrix Factorization. The main contribution of the paper is Multi-Tag influence because they use the simplest and the most used model of Social and Geographical influence. For Multi-Tag influence, the authors first extract the tags for each POI that assigned by each user then use user-rating matrix to create the user-tag matrix. They incorporate the multi-tag, social, and geographical influence with matrix factorization to predix a user i rating to a POI j.

2.6 Multi-Aspect Influence

As contextual information has benfits in POI recommendation, there are some studies that incorporate all of the different contextual information in a single model. For instance, in [30], a multi aspect POI recommendation system has been implemented. The MAPS means that they used Geographical, Temporal, Categorical, and Social influence.Their idea is based on graph and each node corresponds to a location and each user-time tuple considered as an attribute of the location node. For recommendation, they use the Top-Sensitive PageRank model that is an extend of PageRank.

GeoTeCS [31] is a multi-aspect model based on the Weighted Matrix Factorization. In this model, Geographical, Temporal, Categorical, and Social aspect have been considered. The basic idea is according to the [23] and in this paper they just add category and social influence. The

intuition behind the category information is the location with the same category may have potential visits. For the temporal influence, they consider similar check-in time of POIs for the temporal popularity. The user's activity area or influence area is the region that user appearance there (i.e. a POI in the area checked-in by user). Also, a POI influence area is defined as popularity of a POI within a grid.

Regarding the effect of contextual information some of studies incorporate different context to show the impact of them. For example, in [16], the authors try to show the impact of contextual information in recommender systems. They consider successive location prediction as an example to test the prediction accuracy by the different approach. First, they predict by user-user collaborative filtering then they add piece-by-piece the contextual information (temporal, social, and spatial). The results show that the contextual information has a high impact in collaborative filtering and recommender systems.

In [32], the authors addressed the issue of paper [16]. As the paper [16] do not consider the category information and also the different combination of four important contextual information, this paper try to consider them to have better evaluation on the contextual information on POI recommendation.

2.7 Summary

Two main problems in recommender systems, as well as POI recommender systems, are data sparsity and scalability. Initially, researches and studies considered just user-location check-in matrix with memory-based approaches to recommend POIs. These approaches do not address the two main problems very well because they incorporated just user's preference behavior. Also the memory-based approaches suffer from scalability. To address these issues recent studies consider model-based approaches to overcome the scalability problem and the data sparsity, as shown in Table 2.1. They make use of contextual information as additional related information to user's behavior. The scalability of Matrix Factorization method as a model-based approach causes a lot of attention in the recent years. The related works try to solve the sparsity problem by the different combination of the contextual information. Therefore, one of the most important research direction in POI recommendation is how to combine different contextual information based on MF or other methods. Moreover, the modeling of contextual information is another research direction and can be utilized to improve the accuracy of POI recommendation. Table 2.2 demonstrates the summary of related works with the advantage and disadvantage of the related works.

Table 2.1: The summary of contextual information in related works

Related Work	Geographical	Temporal	Social	Category	Content
Xin Liu 2013 [29]				✓	
Jean-Benoit Griesner 2015 [23]		✓			
Ramesh Baral 2016 [30]	✓	✓	✓	✓	
Ramesh Baral 2016 [31]	✓	✓	✓	✓	
Xutao Li 2015 [19]	✓	✓			
Shenglin Zhao 2016 [24]		✓			
Chen Cheng 2016 [20]	✓				
Chen Cheng 2012 [10]	✓		✓		
Zhiyuan Zhang 2018 [11]	✓		✓		✓
Shaojie Qiao 2018 [27]			✓		
Mohammadreza Rahimi 2017 [25]		✓	✓		
Saeid Hosseini 2016 [26]		✓			

Table 2.2: The summary of related works

Related Work	**Method**	**Advantage**	**Disadvantage**
X. Liu 2013 [29]	MF	User's preference transition	Predefined Category Two-Level MF
J. Griesner 2015 [23]	Weighted MF	Spent Time for POI influence	Do not consider Tobler's first law of geography
R. Baral 2016 [30]	PageRank	Multi-aspect	
R. Baral 2016 [31]	Weighted MF	Multi-aspect	
X. Li 2015 [19]	MF	A full user-location matrix	A simple model of geographical The running time
S. Zhao 2016 [24]	Tensor Factorization	Time labeling schema	High-dimensional Matrix
C. Cheng 2016 [20]	MF	Ranking Model Multi-Center Model	Data Sparsity
C. Cheng 2012 [10]	MF	Multi-Center Model	Data Sparsity
Z. Zhang 2018 [11]	MF	New model to incorporate Tag	A simple model for Geographical and Social
S. Qiao 2018 [27]	Memory-based	Different similarity measure	Unscalabality
M. Rahimi 2017 [25]	MF	Combination of time and category	Do not consider geographical on user's behavior
S. Hosseini 2016 [26]	Naive Model	POI Popularity	Do not consider the location of POI

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